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28 November 1979
SUPERSEDING
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MILITARY SPECIFICATION

AMMONIUM NITRATE (UNCOATED)

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of uncoated ammonium nitrate.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.
*

STANDARDS

Military

*	MIL-STD-129	Marking for Shipment and Storage
*	MIL-STD-1188	Commercial Packaging of Supplies and Equipment

Beneficial comments (recommandations, additions, deletions) and any pertinent data which may of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-ESD(R&D), Redstone Arsenal, AL 35809 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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* (Copies of specification, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 128	Method of Analysis of Lubrication Grease

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Code of Federal Regulations

* Title 49 CFR 170-179	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosive and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall meet the requirements of this specification.

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3.2 Chemical analysis. The chemical analysis of the material shall be specified in Table I.

Table I. Chemical Analysis

Constituents	MIN	MAX
Ammonium nitrate (NH_4NO_3), percent	99.0	-
Moisture, percent	-	0.50
Ash, percent	-	0.10
Ether-soluble material, percent	-	0.10
Water insolubles, percent	-	0.10
Soluble chlorides, as NH_4Cl, %	-	0.03
Acidity, as free HNO_3, percent	-	0.02
Acidity, pH	5	7

3.3 Workmanship. The ammonium nitrate shall be uniform in quality, free from foreign materials, and shall be manufactured under conditions and procedures standard in the industry.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of ammonium nitrate.

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The preproduction sample shall be subjected to all the examinations and tests specified herein. Unless otherwise specified, the Government will perform the examinations and tests for preproduction sample acceptance at the contractor's plant. Preproduction samples which do not meet all the requirements of this specification shall be rejected and returned to the contractor. Subsequent quantities will not be considered for acceptance until approval of the preproduction sample has been obtained.

4.3 Lot. A lot shall consist of material produced at one plant with no change in formulation or process. If manufacture is by batch process, each batch shall constitute a lot. A batch shall be defined as in 6.3.

4.4 Acceptance sampling. The number of containers to be chosen at random for acceptance sampling shall be equal to the square root of the total number of containers in the lot. If the number thus obtained is not a whole number, the number of containers to be sampled shall be increased to the next higher whole number. In no case, however, shall the number of containers to be sampled be less than seven (unless there are less than seven containers in the lot, in which case each container shall be sampled).

4.4.1 Primary sample. From each selected container, a sample shall be taken from three or more placed throughout the container. The total weight of the samples taken from each container shall be greater than 50 grams (g). Each sample thus taken shall be mixed thoroughly, placed in a clean dry container, and labeled to identify the material name, original container designation, contract number, and lot number.

4.4.2 Composite sample. Each primary sample shall be subdivided at random to prepare a composite sample (not in excess of 500 g). Primary material not used shall be returned to the primary sample container. After mixing the composite sample thorughly, the composite sample shall be placed in a clean, dry container and sealed. The composite sample shall be identified with the material name, container designation, contract number and lot number. All specified chemical and

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physical tests shall be made on this composite sample representing the lot. Failure of the composite sample to pass all of the tests herein shall result in rejection of the lot represented.

4.5 Classification of tests. Inspection and testing of ammonium nitrate shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.6 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all tests specified herein.

4.7 Quality conformance tests. Quality conformance tests for acceptance of the ammonium nitrate shall consist of the following tests:

Constituents	Test
Ammonium nitrate	4.8.1
Moisture	4.8.2
Ash	4.8.3
Ether - Soluble material	4.8.4
Water insolubles	4.8.5
Soluble chloride, as NH_4Cl	4.8.6
Acidity, as free HNO_3	4.8.7
Acidity, pH	4.8.8

4.8 Acceptance tests. The following procedures shall be used to determine that the requirements of this document have been met. Any change in test procedures employed shall receive prior approval of the procuring activity. Unless otherwise specified, all tests shall be run in duplicate. The average of the two results shall be taken as the test result.

4.8.1 Ammonium nitrate (NH_4NO_3). Add 25 milliliters (ml) of approximately 40 percent formaldehyde solution and a few drops of phenolphthalein indicator solution to 100 ml of freshly boiled distilled water in a 125-ml Erlenmeyer flask. Neutralize the solution with 0.15N sodium hydroxide (NaOH), add a weighed portion of approximately 1 g of sample (weighed to the nearest 0.01 g), heat mixture to 60 degrees Celsius (C) (140 degrees Fahrenheit (F); then cool to room temperature. Titrate the solution with 0.5N NaOH to an end point which persists at least 30 seconds.

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Percent ammonium nitrate equals $\frac{8.005\ AB \text{ minus } 1.49C}{W}$

Where: A equals volume of sodium hydroxide used, ml.
B equals normality (N) of sodium hydroxide solution.
C equals ammonium chloride in sample (4.8.6), percent.
W equals weight of sample taken, g.

4.8.1.1 Acceptance criteria. For the lot represented to pass the ammonium nitrate test, the value obtained for percent ammonium nitrate shall be not less than the value specified in 3.2.

4.8.2 Moisture determination.

4.8.2.1 Apparatus. The apparatus used for determination of moisture content of the sample shall be Aquameter, Model KF-2 or KF-3, Beckman Instruments, Inc., Fullerton, CA, or approved equivalent. The Aquameter shall be prepared for operation as described in the technical manual furnished by the manufacturer (Beckman Instruments, Inc.). Use of an alternate item of equipment approved by the procuring activity will necessitate use of the specific technical manual prepared by the manufacturer.

4.8.2.2 Reagents.

a. Karl Fischer reagent. Karl Fischer reagent must have a strength such that each milliliter of Karl Fischer reagent corresponds to 0.0014-0.0023 g of water. Dilute 750 ml of commercially available established Karl Fischer reagent (with water equivalent of 0.005-0.007 g/ml) to 2000 ml with absolute methanol (0.1 percent water, maximum). Mix well and allow to stand overnight before use. Determine the water equivalent (A) of this solution as follows:

1. Use sodium tartrate dihydrate ($NO_2C_4H_40_6.2H_20$) as a primary standard (with a water content of 15.66 percent) for standardizing Karl Fischer reagent. If the water content value is in question, it may be determined by heating some of the salt at 150 degrees C (302 degrees F) for 3 hours. Should the value (as determined)

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differ from the theoretical value of 15.66, then the experimental value shall be used in the determination of water equivalent (A) of the Karl Fischer reagent; instead of the 15.66 in the formula below, the factor should be 10P, where P is percentage moisture (as determined). Rapidly transfer 0.090-0.110 g weighed to the nearest 0.0001 g) of reagent-grade $Na_2C_4H_4O_6.2H_2O$ to the titration vessel.

2. Titrate to an end point in the same manner as with the sample. (See 4.8.2.3)

3. Repeat the standardization procedure until three successive results agree within five parts per thousand.

* 4. If the indicated water equivalent (A) of the Karl Fischer reagent is less than 0.0014 g of water per ml Karl Fischer reagent, it may be due to the presence of too much water in the absolute methanol used. In this case, distill the methanol from metallic calcium or calcium hydride. Passing the methanol through a column of Molecular Sieves, Type 4A, may also reduce the water content of the methanol sufficiently.

Water equivalent (A) equals $\frac{0.1566}{V} W$

Where: A equals water equivalent of the Karl Fischer reagent, g/ml.
W equals weight of $Na_2C_4H_4O_6.2H_2O$ taken, g.
V equals volume of Karl Fischer reagent used, ml.

b. Water-in-methanol solution. The water-in-methanol solution should contain 0.0015-0.0020 g of water per ml of solution. A good grade of commercial absolute methanol contains approximately 0.0010 g of water per ml of methanol. Water content can be adjusted by adding 1.0 g of water to 1000 ml of the water-in-methanol solution to produce a change of 0.0010 g per ml. Determine the relative strength of the water-in-methanol solution in terms of Karl Fischer reagent as follows:

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1. Put about 50 ml of the anhydrous methanol used in 4.8.2.2a into the titration breaker of the Aquameter. Add a slight excess of Karl Fischer reagent (4.8.2.2a), then back-titrate with water-in-methanol solution (4.8.2.2b). Then run in an additional 5 to 8 ml of Karl Fischer reagent, read to the nearest 0.01 ml, and again back-titrate with water-in-methanol solution (read to the nearest 0.01 ml). Repeat the addition and back-titerating steps twice more to provide triplicate determinations of the equivalency ratio. Calculate the ratio (B) of the Karl Fischer reagent to that of the water-in-methanol solution. The range of the ratios calculated from the three titrations should not be greater than 0.04. If the range exceeds 0.04, continue making titrations until three ratios are obtained whose range does not exceed 0.04. Then determine the average ratio from all the ratios which have been obtained.

4.8.2.3 Procedure. Determine the moisture content of the sample by the Karl Fischer method using a direct-titration technique. Introduce the titration beaker, through the opening in the diaphragm, approximately 10 g of sample weighed to the nearest 0.001 g. Close the opening, start the stirrer and press the titrate button to titrate with Karl Fischer reagent (4.8.2.2a). When the indicator light glows, read the Karl Fischer buret to the nearest 0.01 ml. Where necessary, water methanol solution (4.8.2.2b) may be used to back titrate.

$$\text{Percent moisture equals } \frac{100A\,(V_{KF} \text{ minus } BV_{WM})}{W}$$

Where: A equals weight of water equivalent to 1.00 ml of Karl Fischer reagent, g/ml.

V_{KF} equals volume of Karl Fischer reagent titrant used, ml.

V_{WM} equals volume of water-in-methanol solution titrant used, ml.

B equals ratio of Karl Fischer reagent to that of water-in-methanol solution, ml/ml.

W equals weight of sample taken, g.

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4.8.2.4 Acceptance criteria. For the lot represented to pass the moisture test, the value obtained for percent moisture shall be no greater than the value specified in 3.2.

* 4.8.3 Ash. Determine the amount of ash utilizing the procedure given by ASTM D 128.

4.8.3.1 Acceptance criteria. For the lot represented to pass the ash test, the value obtained for percent ash shall be no greater than the value specified in 3.2.

4.8.4 Ether-soluble material. Dry a clean, 150-ml beaker in an oven at 100 plus or minus 5 degrees C (212 plus or minus 9 degrees F) for approximately one hour, cool in a desiccator, and weigh to the nearest 0.001 g. Weight to the nearest 0.001 g approximately 25 g of the specimen and transfer to the thimble of a Soxhlet extractor containing anhydrous ether. Extract the specimen for approximately 2 hours. Save the extracted specimen for the water-insoluble determination (4.8.5). Evaporate the ether extract in the extraction flask to a volume of approximately 50 ml on a steam bath and transfer to the weighed beaker. Rinse out the extraction flask with several 5-ml portions of ether and add them to the main extract in the beaker. Cover the beaker with a ribbed watch glass and evaporate on a steam bath. After the solvent is removed, heat the beaker and residue in an oven at 100 degrees C (212 degrees F) for one hour, cool in a desiccator, and weigh to the nearest 0.001 g.

$$\text{Percent ether-soluble material equals } \frac{100A}{W}$$

Where: A equals weight of residue from sample, g.
W equals weight of sample taken, g.

4.8.4.1 Acceptance criteria. For the lot represented to pass the ether-soluble-material test, the value obtained for percent ether-material shall be no greater than the value specified in 3.2.

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4.8.5 Water insolubles. Using the ether-extracted specimen from 4.8.4, dissolve it in approximately 100 ml of distilled water. Weigh a dry, sintered-glass funnel to the nearest 0.001 g, then filter the mixture carefully through the funnel in order that all undissolved residue may be collected. Dry the funnel and residue for one hour in an oven at 105 plus or minus 5 degrees C (221 plus or minus 9 degrees F), allow to cool to room temperature in a desiccator, then reweigh to the nearest 0.001 g.

$$\text{Percent water insolubles equals } \frac{(A \text{ minus } B)\ 100}{W}$$

Where: A equals weight of funnel plus residue, g.
B equals weight of funnel empty, g.
W equals weight of sample taken in 4.8.4, g.

4.8.5.1 Acceptance criteria. For the lot represented to pass the water-insolubles test, the value obtained for percent water insolubles shall be no greater than the value specified in 3.2.

4.8.6 Soluble chlorides, as NH_4Cl. Place approximately 50 ml of chloride-free distilled water in a porcelain evaporating dish. Add 5 g of sample (weighed to the nearest 0.0001 g). Acidify the solution with approximately 2 ml of 3N nitric acid. Titrate the solution with an excess of 0.1N silver nitrate. Add a few drops of ferric alum indicator solution and, while swirling, titrate the excess silver nitrate with 0.1N ammonium thiocyanate. The end point is taken as the first permanent pink or red color observed.

$$\text{Percent ammonium chloride equals } \frac{(AB \text{ minus } CD)\ 5.35}{W}$$

Where: A equals volume of silver nitrate used, ml.
B equals normality of silver nitrate solution.
C equals volume of ammonium thiocyanate solution used, ml.
D equals normality of ammonium thiocyanate solution.
W equals weight of sample taken, g.

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4.8.6.1 Acceptance criteria. For the lot represented to pass the soluble-chloride test, the value obtained for percent NH_4Cl shall be no greater than the value specified in 3.2.

4.8.7 Acidity, as free HNO_3. Dissolve approximately 100 g of sample (weighed to the nearest 0.01 g) in 400 ml of distilled water. Filter, add methyl red indicator, and titrate with 0.1N sodium hydroxide (NaOH). Calculate the acidity as follows:

$$\text{Acidity, as percent nitric acid equals } \frac{6.3\ N\ V}{W}$$

Where: N equals normality of NaOH solution.
V equals volume of NaOH solution used, ml.
w equals weight of sample taken, g.

4.8.7.1 Acceptance criteria. For the lot represented to pass the acidity test, the value obtained for percent free HNO_3 shall be no greater than the value specified in 3.2.

4.8.8 Acidity, pH. The acidity, pH, specified in 3.2 shall be determined in accordance with 4.8.8.2.

4.8.8.1 Apparatus. Electrometric pH apparatus, or equal.

4.8.8.2 Procedure. Dissolve approximately one g of sample (weighed to the nearest 0.01 g) in 100 ml of freshly boiled distilled water contained in a 200-ml beaker. Determine the pH of the solution electrometrically using a pH meter.

4.8.8.3 Acceptance criteria. For the lot represented to pass the acidity (pH) test, the value obtained shall be within the range shown in 3.2.

* 5. PACKAGING

* 5.1 Preservation, packing, unitization, and marking. Commercial packaging (see 6.2).

* 5.1.1 Commercial. Preservation, packing, unitization, and marking shall be in accordance with MIL-STD-1188, and also provisions of Title 49 of Code of Federal Regulations, Parts 170-179.

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6. NOTES

6.1 Intended use. The ammonium nitrate described in this specification is intended for use as an ingredient in ammonium-nitrate-base solid propellants.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of specification.

b. Whether a preproduction sample is required (see 3.1).

* c. Type and size of shipping container (see 5.1).

6.3 Definition.

6.3.1 Batch. A batch is defined as that quantity of material which has been subjected to one or more chemical or physical processes (or combinations thereof) intended to produce a desired product having substantially uniform characteristics.

6.4. Safety and health warning. Ammonium nitrate is an oxidizing material. When contaminated with organic materials, it is sensitive and may explode when subjected to friction or fire.

* 6.5 Changes to previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
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DOCUMENT IDENTIFIER AND TITLE
MIL-A-47240A(MI) AMMONIUM NITRATE (UNCOATED)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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